

Analysis of Transistor-level CMOS Inverter in LTspice

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1 Design Parameters & Tool Setup

1.1 Parameters

The NMOS and PMOS transistors are modeled using the typical values for 180nm CMOS Small-Signal Model [1]. The SPICE directives for the NMOS and PMOS transistors are as follows:

- **NMOS:** `.model NMOS_180nm NMOS (LEVEL=1 VTO=0.8 KP=90u GAMMA=0.5 PHI=0.68 LAMBDA=0.06 TOX=0.02u CJ=2.4e-4 CJSW=2.0e-10 CGSO=2.0e-10 CGDO=2.0e-10 PB=0.9 NSUB=6e15)`
- **PMOS:** `.model PMOS_180nm PMOS (LEVEL=1 VTO=-0.9 KP=30u GAMMA=0.5 PHI=0.68 LAMBDA=0.06 TOX=0.02u CJ=2.4e-4 CJSW=2.0e-10 CGSO=2.0e-10 CGDO=2.0e-10 PB=0.9 NSUB=6e15)`

The Length (L) of both transistors is set to $0.18\mu m$, and the Width (W) is set to $1\mu m$ for the NMOS and $2\mu m$ for the PMOS.

1.2 LTspice Schematic

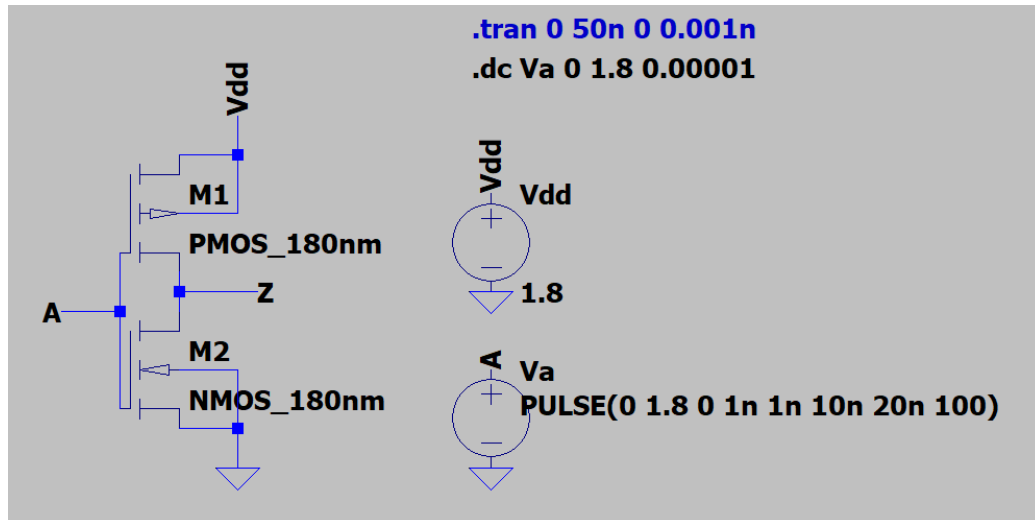


Figure 1: Circuit setup of the transistor-level CMOS inverter in LTspice.

2 DC Analysis (Voltage Transfer Characteristics)

2.1 VTC Curve Generation

To obtain the Voltage Transfer Characteristic (VTC), a DC sweep simulation is performed, sweeping the input voltage from 0V to 1.8V.

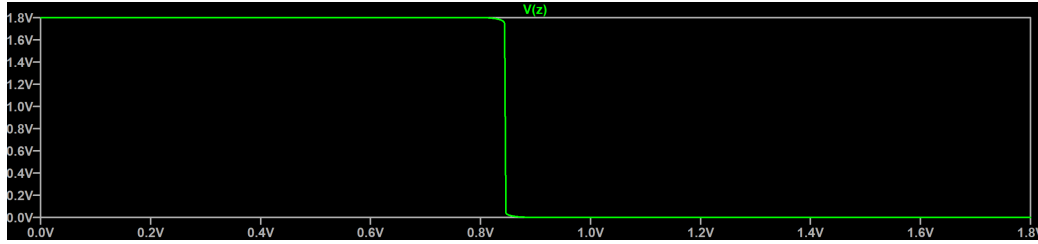


Figure 2: Voltage Transfer Characteristic (VTC) of the CMOS Inverter.

From Fig.2, the relationship between the V_a and V_z is shown. It was found that when V_a was lower than 0.8V, the output voltage (V_z) was high, and when V_a was higher than 0.9V, V_z was low.

2.2 Critical Voltages

From the VTC curve, the critical voltages are extracted (where the slope is -1):

- $V_{OH} = 1.8 \text{ V}$
- $V_{OL} = 0 \text{ V}$
- $V_{IH} = 0.857 \text{ V}$
- $V_{IL} = 0.829 \text{ V}$

2.3 Noise Margins

The noise margins are calculated using the extracted critical voltages:

- High Noise Margin ($NM_H = V_{OH} - V_{IH}$) = 0.943 V
- Low Noise Margin ($NM_L = V_{IL} - V_{OL}$) = 0.829 V

2.4 Switching Threshold

The switching threshold (V_M), where $V_{in} = V_{out}$, is measured from the VTC:

- $V_M = 0.845 \text{ V}$

3 Transient Analysis (Timing Characteristics)

3.1 Transient Simulation Setup

To analyze the timing characteristics, a transient simulation is performed. The input is driven by a pulse voltage source with the following parameters:

- $V_{initial} = 0$ V, $V_{on} = 1.8$ V
- $T_{delay} = 0$ s, $T_{rise} = 1$ ns, $T_{fall} = 1$ ns
- $T_{on} = 10$ ns, $T_{period} = 20$ ns

3.2 Propagation Delay and Timing

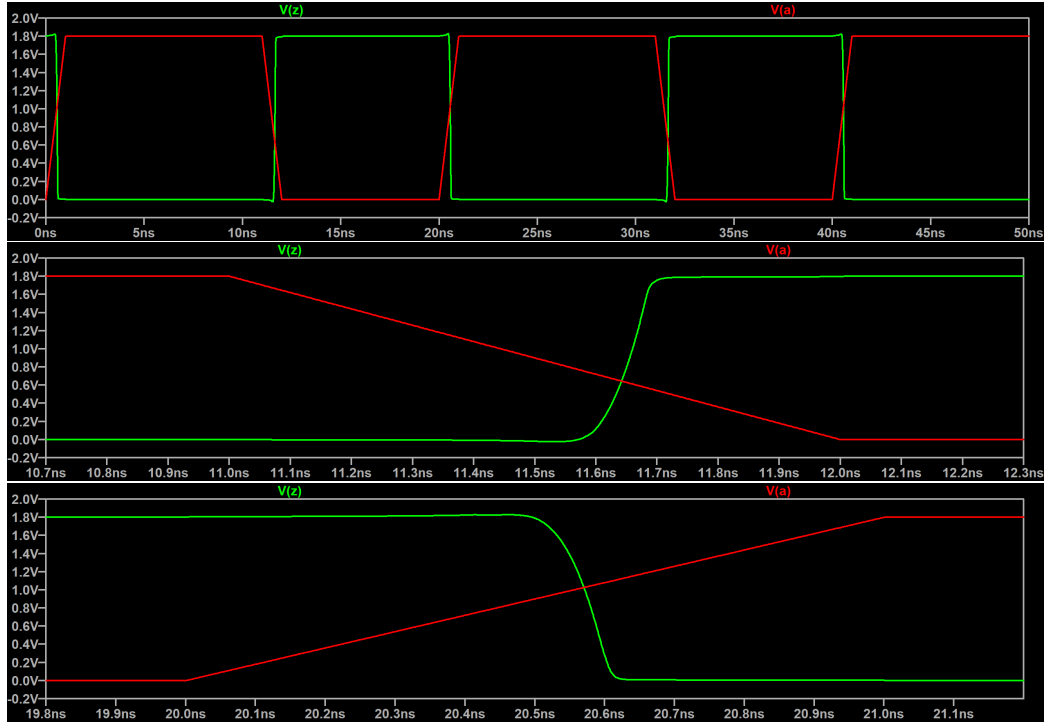


Figure 3: Top: Transient response of the CMOS inverter. Middle: Zoomed-in view of the rise time. Bottom: Zoomed-in view of the fall time.

The propagation delay is measured between the 50% points of the input and output transitions:

- High-to-Low Delay (t_{pHL}) = 7.5 ps
- Low-to-High Delay (t_{pLH}) = 150 ps
- Average Delay (t_p) = 79 ps

The rise and fall times are measured between the 10% and 90% points of the output signal:

- Rise Time (t_r) = 74 ps
- Fall Time (t_f) = 72 ps

References

- [1] Scribd. 180nm process typical parameter values.
<https://www.scribd.com/presentation/470685433/180nm-process-typical-parameter-values>.